

INTERNSHIP REPORT



Implementation of a Factsheet Reporting Project

*The creation of reports for a financial product
for banks and fund managers*

Irmak Damla Özdemir

Duration:	10. Feb 2026 - 17. Jul 2026
Internship Coordinator:	Prof. Dr.-Ing. Erik Schaffernicht
Project Manager:	Sebastian Kempf
Student ID:	5123127
E-mail:	irmakozfecontact@gmail.com
Phone:	0049 1575 462 59 32
Semester:	Summer Semester 2026
Field of Study:	B. Eng. Computer Science
University:	Technical University of Applied Sciences Würzburg-Schweinfurt
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Abstract

This report presents a comprehensive analysis of my internship experience at Anevis Solutions, a leading reporting provider. The primary focus of the internship was to understand and implement IT-based automated reporting solutions for the financial sector. tbd

The internship provided an invaluable opportunity to apply theoretical knowledge acquired in academic students to real-world scenarios. I participated in several projects, notably the "Christian Hintz Vermögensverwaltung AI Leaders Factsheet Reporting", which aimed to provide individual and dynamic reporting for their fund investment class.

Additionally, the experience offered a better understanding of Linux systems, Bash scripts, database management and charting libraries like JFreeChart. I implemented various so called parsers to contribute automation of reportings. This Parser types were used to take the data from customer as any file type and save it consistently to the databases and then fetch the data from the database and create charts and analytics for reportings via Jaspersoft Studio.

The intership underscored the importance of adaptability, teamwork and continuous learning. It also highlighted the critical role of clean and reusable implementation for the maintainability of distributed systems.(?) Furthermore, it reinforced my passion to develop sustainable, reusable and customized software for IT Technologies and has shaped my career aspirations in Backend Development.

This report details the project undertaken, skills developed and lessons learned throughout the internship, within a very organized work culture and with constant feedbacks from colleagues and supervisor.

Abstract

Dieser Bericht präsentiert eine umfassende Analyse meines Praktikums bei Anevis Solutions, einem Anbieter im Bereich Reporting-Lösungen. Der Schwerpunkt des Praktikums lag auf dem Verständnis und der Implementierung IT-basierter automatisierter Reporting-Lösungen für den Finanzsektor.

Das Praktikum bot eine wertvolle Gelegenheit, theoretisch erworbenes Wissen aus dem Studium in realen Anwendungen einzusetzen. Dabei arbeitete ich an mehreren Projekten mit, insbesondere am Projekt „Christian Hintz Vermögensverwaltung AI Leaders Factsheet Reporting“, das darauf abzielte, individuelle und dynamische Berichte für eine bestimmte Investmentfonds-Klasse bereitzustellen.

Darüber hinaus ermöglichte die Tätigkeit ein vertieftes Verständnis von Linux-Systemen, Bash-Skripten, Datenbankmanagement sowie Charting-Bibliotheken wie JFreeChart. Im Rahmen der Automatisierung von Reporting-Prozessen implementierte ich verschiedene Parser-Typen. Diese dienten dazu, Kundendaten in unterschiedlichen Dateiformaten zu verarbeiten, konsistent in Datenbanken zu speichern und anschließend für die Erstellung von Diagrammen und Analysen in Jaspersoft Studio wieder abzurufen.

Das Praktikum verdeutlichte die Bedeutung von Anpassungsfähigkeit, Teamarbeit und kontinuierlichem Lernen. Zudem wurde die Relevanz sauberer und wiederverwendbarer Implementierungen für die Wartbarkeit verteilter Systeme hervorgehoben. Diese Erfahrungen stärkten mein Interesse an der Entwicklung nachhaltiger, wiederverwendbarer und maßgeschneiderter Softwarelösungen im IT-Bereich und prägten meine beruflichen Ziele in der Backend-Entwicklung.

Dieser Bericht dokumentiert die durchgeführten Projekte, die erworbenen Fähigkeiten sowie die gewonnenen Erkenntnisse während des Praktikums in einem strukturierten Arbeitsumfeld mit kontinuierlichem Feedback durch Kollegen und Vorgesetzte.

Approval

Approved By

Company Supervisor

Name

Signature

Date

Academic Supervisor

Name

Signature

Date _____

Confidentiality Statement

I hereby confirm that confidential information obtained has been handled appropriately and has not been disclosed without permission.

Würzburg, den XX July 2026

.....
Sebastian Kempf / responsible person

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1 Introduction

Computer science has become one of the most important fields in today's world due to the rapid digital transformation and the strong technology sector in Germany. Software development and information technologies play an essential role across almost every industry. As a result, computer engineers are expected not only to have solid theoretical knowledge but also practical experience in real-world applications.

For me, applying the theoretical knowledge gained during my studies to real projects has always been of great importance. Germany's practice-oriented education system and the opportunities offered by THWS were key factors in my decision to study here. Through my internship, I had the opportunity to further develop my technical skills as well as gain valuable insight into the professional working environment in Germany.

1.1 Motivation

During my search for a suitable internship, several factors played an important role in my decision-making process. Gaining technical experience, working within a team under a structured project environment, and improving my personal weaknesses were key criteria for selecting my internship placement. In particular, I aimed to deepen my knowledge of a programming language and further develop my technical skill set.

One aspect that particularly stood out to me at Anevis was the structured way of working, including regular meetings, clearly defined task distribution, and the opportunity to independently complete assigned responsibilities. Another important motivation for me was the chance to adapt to and work within an existing software system. This was especially relevant, as I recognized that integrating into an already established codebase is more challenging than developing a project from scratch.

Overall, this experience represented an important step for me in improving my ability to work in a team, adapt to existing systems, and develop a more disciplined and structured approach to software development.

2 Company Overview

2.1 Description

Anevis Solutions GmbH is a software company based in Würzburg, Germany, specializing in digital reporting solutions for the financial industry across Europe. The company is a leading reporting provider, offering software and services that support banks, asset managers, fund providers, and other financial institutions in automating and optimizing their reporting processes while reducing manual effort.

2.2 Products

2.2.1 Marketing Suite

Factsheets: Automated generation of standardised fund and portfolio factsheets based on up-to-date financial data.

Institutional Reporting: Creation of detailed reports for institutional investors including performance and portfolio information.

Sales Presentations: Generation of customised presentation materials for client communication and sales activities.

ESG Reporting: Production of sustainability-related investment reports based on ESG data.

Website Widgets: Interactive components displaying live fund data and performance metrics on websites or client portals.

Live Hosting for Documents: Digital distribution and hosting of dynamically updated reports and marketing materials.

2.2.2 Regulatory Suite

UCITS KIID: Automated generation of Key Investor Information Documents for UCITS funds.

PRIIPs KID / MiFID II: Standardised regulatory disclosure documents for EU compliance requirements.

FIDLEG BIB: Regulatory reporting solution aligned with Swiss financial regulation.

SFDR Reporting: Sustainability disclosure reporting in accordance with SFDR requirements.

PAI and EET Reporting: Reporting of Principal Adverse Impacts and ESG template data.

AIFMD Annex IV Reporting: Regulatory reporting for Alternative Investment Fund Managers.

Solvency II TPT: Reporting solution for insurance sector regulatory requirements.

2.2.3 Data Management Suite

Datawarehouse: Centralised storage and management of financial and market data.

Data Integration: Integration and harmonisation of multiple internal and external data sources.

Data Validation: Ensures consistency, accuracy, and quality of input data.

2.2.4 Risk & Compliance Suite

Risk Management Service: Portfolio risk analysis and exposure monitoring.

Transaction Cost Calculation: Calculation of transaction costs for performance and regulatory reporting.

Market Conformity Check / Best Execution: Analysis of trade execution quality and market conformity.

Valuation & Pricing Service: Automated valuation and pricing of financial instruments.

2.3 Departments

2.3.1 Sales & Marketing

Satis ve pazarlama departmani mevcut musterilerle ilgilenirken ayni zamanda yeni musteriler kazanmakla sorumlulardir. Bunun yaninda muhasebe ve finans yönetimi bu departman tarafından yapilir.

2.3.2 Human Resources

Human Resources oncelikli olarak sirketin surekli buyumesi dogrultusunda acik pozisyonlarin nitelikli personellerle doldurulmasini saglar. Bu kapsamda is ilanlari hazirlanir, is gorusmeleri

yapılır ve sözleşmeler imzalanır. Ek olarak şirketin sosyal medya hesaplarının yönetimi, içerik hazırlanması ve event management da bu birimin görevidir.

2.3.3 Operations

Operasyonlar bölümü, birbirleriyle yakın iş birliği içinde çalışan Proje Yönetimi ve Operasyon Geliştirme (Operations Development) alt birimlerinden oluşur. Bu iki birim, müşteri projelerinin hayata geçirilmesinden ortaklaşa sorumludur. Ayrıca, hem teknik hem de organizasyonel düzeyde birinci seviye destek (First-Level Support) hizmetleri de bu bölümlerin sorumluluk alanına girer.

Project Management (PM)

Proje yönetimi alt birimi, müşteri projelerindeki gerekliliklerini takip etmekle müşteri ve developer arasındaki köprüyü kurmakla sorumludur. Müşterinin birincil iletişim noktasıdır, müşteriye danışmanlık yapar. Tamamlanan projelerde ise müşterilerin feedbackini ve update'leri takip eder.

Operations-Development (OPSDEV)

OPSDEV alt birimi müşteri projelerinin teknik implementasyonundan sorumludur. Çözümler geliştirir ve değişiklik taleplerini uygular. Mission is to take care of the technical implementation of projects and technical support for OPS. Böylece departman, ürünlerin tamamen otomatize edilmesinde rol oynar. Bölümün diğer faaliyetleri arasında OPS-DEV süreçlerinin geliştirilmesi ve optimize edilmesi yer alır. Ayrıca OPS-DEV, teknik konularda Proje Yönetimi ekibine destek sağlar.

Development (DEV)

Yazılım Geliştirme bölümü, özellikle BT altyapısının ve şirket bünyesindeki yazılım mimarisinin teknik bakımından ve işleyişinden sorumludur. Amaç, BT sistemlerinin kesintisiz ve sorunsuz çalışmasını sağlamaktır.

Başlıca görevler arasında dahili araçların ve kod kütüphanelerinin geliştirilmesi, veri tabanlarının ve sunucuların bakımı yer almaktadır.

Bölüm ayrıca, teknik sorunlar ortaya çıktığında etkili çözümler geliştirmek amacıyla Operasyonlar departmanlarına teknik destek sağlamaktadır.

Quality Assurance (QA)

Team QA implemente edilen Taskleri local test etmekte ve requirementların karşılandığından emin olmakla sorumludur

2.4 TechStack and Tools



Figure 2.1: Technologies Used in the Company

2.5 Team Events

2.5.1 Student Event SS 26

An evening with all students (working students and interns), including short presentations, mini-games, and a shared dinner.

2.5.2 OPS Team Event

An evening in Steinbachtal, Würzburg. We had an archery session and competed in teams. Afterwards, we had dinner and desserts together at Schützenhof.

3 Tasks in the Company

The internship includes a broad range of side tasks in addition to the main internship project, providing real-world working experience and insight into office culture.

3.1 Meetings

3.1.1 OPSDEV Daily Stand-up

Every morning at 09:05, a short meeting is held via Google Meet where team members share updates on their current work and discuss ongoing tasks.

3.1.2 OPSDEV Check-in Meetings

Biweekly development team check-in meetings focusing on progress, challenges, concerns, achievements, and overall well-being, with all discussions recorded in the OPSDEV meeting protocols.

3.1.3 OPS Review / Consultation Meetings

Regular OPS team sessions where team members can ask questions, discuss issues, and receive support regarding Anevis products.

3.1.4 Quarterly Reviews

Quarterly review meetings are held where each team presents its current progress, key achievements, and ongoing activities in a structured, sequential manner. The meetings serve to provide an overview of the overall project status across all teams.

3.1.5 Internship Feedback Sessions

Feedback sessions were conducted at regular intervals throughout the internship to exchange mutual feedback, review overall progress, and discuss expectations and satisfaction levels from both sides.

3.2 Onboarding

3.2.1 Onboarding Presentation from HR

The onboarding presentation provided an introduction to Anevis and covered its history, organizational structure, office environment, and company culture. It also included information about meetings, team events, internal communication practices, as well as an overview of the company's products, tools, and working guidelines.

3.2.2 OPSDEV Training Exercises

- *Exercise Streams:* A set of exercises designed to strengthen proficiency with Java Streams, reflecting their extensive use within the internal system and covering common stream operations and data processing patterns.
- *Exercise Databases:* Practical exercises focused on gaining experience with MongoDB and PostgreSQL, which are the primary databases used within the internal system, including querying, data modelling, and basic database operations.
- *Exercise Spring JPA:* An exercise focused on Spring, Hibernate, JPA, SQL, and Java. The task involved generating pie and ring charts with JFreeChart, exporting them to PDF, applying clean OO design principles, and supporting the solution with unit tests, streams, and lambdas.
- *Exercise Factsheet Project:* A practical factsheet generation exercise simulating a syntetic client project. The task involved working with the internal repositories like document engine, setup scripts; and Git workflows, real databases, JFreeChart reporting components to produce a complete factsheet in a dedicated development branch.

3.3 OPSDEV - Daily Development Tasks

OPSDEVBOX Tasks

OPSDEVBOX taskleri automatisierungta onemli rol oynayan sirketin kendi internal sistemi icin implemente edilen tasklerdir. Ornegin musteriden gelen datalarin veri bankasina nasil kaydedilecegi, database structurelari update edilmesi yani repositorylerdeki genel duzenlemeler bu task basligi altinda yapilir

Task	Description
OPSDEVBOX-651	Add region ultimo mapping to the project setup
OPSDEVBOX-678	Make category mandatory in project setup parser
OPSDEVBOX-679	Fix marketing classifications with empty identifiers
OPSDEVBOX-662	Move series from definitions data to series mappings
**LLBSWISS-255	Implement parser to insert valor

Table 3.1: OPSDEVBOX Tasks Overview

BANTLEON Fast Charting Project

Fast Charting Projesi, Bantleon sirketine ait esg report projesi için ops team tarafından hızlı ve efektif bir şekilde chart üretimini sağlayan databasele bağlantısı olmayan pratik yardımcı bir tool olarak geliştirilmiştir. Özetle bu class ile chartlar customization ile dilediği şekilde visualize edilebilir.

İşte bazı chart örnekleri:

Task	Description
BANTLEON-249	fast-charting-for-bantleon-esg-report

Monitoring-Reporter

Sirkette internal sistem kontrolünü sağlamak ve bunları dilediği aralıklarla mailde raporlaştırmak için bir servis bulunmaktadır ve burada ihtiyaca göre düzenli raporlar üretilebilmektedir. Burada benim görevim, şirketin databaseinde bulunan project componenti olmayan projeleri saptamak ve onların bilgisini bir tablo halinde mail olarak yönlendirmektir. Burada internal sistemde zaten hali hazırda var olan bir Class i extend ederek customize edip bir reporter implemente ettim

Task	Description
OPSDEVBOX-680	monitor-dead-projects

Customer-specific Tasks

Customer specific tasklar, pm in müşterilerden aldıkları geri bildirim ve taleplere yönelik oluşturulan çeşitli tasklardır. Bunlara örnek olarak bir müşterinin adının, product adının

degistirilmesi, chartların legendinin 2 sütunlu yapılması, proje componenti silinmesi, static data update edilmesi, basketlerin ve pricelerin update edilmesi, vb. Özetle internship projecte oranla kapsami çok daha küçük olan yan taskler.

3.4 Additional

Confluence Article about JFreeCharts

ISO 27001 Certification Training

General Data Protection Awareness Training

Anecon Support

The perfect opportunity for networking in the financial service industry Yilda bir kere duzenlenen, cesitli banka ve fon sirketlerinden konusmacilarinin oldugu, ... 18 Haziran 2026 dan gerceklesen bu konferansta city helper olarak gorev aldim

Cleaning Service (maybe)

3.5 Given Materials

The following books were used as reference material during the internship:

- Robert C. Martin: *Clean Code: A Handbook of Agile Software Craftsmanship*
- Raoul-Gabriel Urma, Mario Fusco, Alan Mycroft: *Java 8 in Action*

4 Internship Project

4.1 Customer Overview

Christian Hintz Vermögensverwaltung GmbH is an independent German asset management company based in Stuttgart. The firm specializes in providing discretionary wealth management services for private investors and businesses, focusing on individually tailored investment strategies and portfolio management. Operating as a regulated financial institution under the supervision of the German Federal Financial Supervisory Authority (BaFin), the company offers services such as strategic portfolio management, financial planning, and investment advisory with a strong emphasis on long-term, goal-oriented wealth building.

4.2 Customer KVG

A Capital Management Company is a regulated financial institution that manages investment funds for investors. Its main tasks include portfolio management, risk management, and ensuring compliance with regulations. It also makes sure that investment strategies are followed according to the fund's goals. In many cases, it acts as a link between asset managers, custodians, and other financial service providers, while being supervised by the relevant financial authority in its country.

In Germany, this type of institution is called a KVG (Kapitalverwaltungsgesellschaft) and is regulated under the German Capital Investment Code (KAGB) by BaFin.

In this project, the client's Capital Management Company is Monega Kapitalanlagegesellschaft mbH. Monega is a German investment management company that provides fund administration and asset management services. In this context, Monega acts as the data provider for client investment information, making sure that structured and validated financial data is available for processing and reporting.

4.3 Preparation

Proje süreci boyunca müşteri ve yazılımcı arasında köprü görevi kuran project manager ile çalıştım. Project manager müşteri ile birlikte proje için requirementları belirledi ve

Jira Taskte bunlari listeledi. Ben de bu sayede bir roadmap olusturabildim. Musteriden gelen Input verilerinin kaydedilebilmesi icin once sirketin internal sisteminde bulunan database scripts reposunda setup scripts dosyalarini olusturacaktim sonrasinda yine internal sistemdeki document engine de parser implemente edecektim ve en son yine burada otomatize factsheet olusumu icin parser implemente edecektim. Tüm bu plan icin version control sistemi Git ile repositorylere projeye uygun branchler olusturarak baslangic yaptim.

4.4 Requirements

The scope of the project covers the generation of factsheets for the AI Leaders fund. The fund is structured into two share classes, AI Leaders A and AI Leaders C. Within this project, the AI Leaders C investment class has been implemented first, serving as a base template for the AI Leaders A class as well. This design allows both share classes to be generated using the same template with different setup scripts, ensuring reusability and consistency. The main characteristics of the project can be summarized as follows:

- Factsheets can be generated and delivered at configurable time intervals based on future client requirements.
- The existence of two share classes is due to differences in fees and expense structures, which may result in different performance outcomes for the same fund.
- Input data is provided by the client's KVG in XLSX format and is processed through the implemented application before being stored in a database.
- Project requirements, including sample data files, are shared via Jira as attachments by the project manager for testing purposes.
- The project manager defines the factsheet structure, including statistics, charts, and logos, in collaboration with the client and provides these requirements for implementation in the parser.
- Chart styling elements such as color palettes, value ranges, and text annotations are predefined during the design phase.

Jaspersoft is a flexible reporting and analytics platform used to transform raw data into structured and readable reports. It retrieves both static and dynamic data from a configured database, enabling the creation of interactive dashboards, visual charts, and PDF reports.

4 Internship Project

Overall, factsheets can be generated at any time using the stored database information in combination with the configured Jaspersoft template.

charts:

- Performance Chart, rollierende volatility chart, max drawdown chart
- asset allocation, country allocation, sector allocation, currency allocation ring chart
- Top 10 security positionen bar chart

statistics:

- monthly and yearly wertentwicklung statistics
- risikokenzahlen for year 1

The tables below represent the column structure of the XLSX files provided by the client. Data is extracted and processed based on these defined columns:

A	E	F	G	H	S	V
AS_OF_DATE	NAME	WKN	ISIN	CURRENCY	FUNDVOLUME	PRICE

Table 4.1: FondsKompakt Input Column Mapping

AA	AB
DIVIDEND	DIVIDEND_DATE

Table 4.2: FondsKompakt Input Column Mapping

E	F	G	I	P	W	V
ISIN	SECURITY NAME	ASSET	CURRENCY	MARKET VALUE	ISSUER	SECTOR

Table 4.3: Bestand Input Column Mapping

AC
COUNTRY

Table 4.4: Bestand Input Column Mapping

4.5 Implementation

4.5.1 Setup Scripts

Musterilerden gelen veriler databankalarına direkt manuel olarak eklenmemektedir. Bunun için bir database-scripts adında bir repo bulunur ve setup scripleri sayesinde data manipulation uygulanabilmektedir. Bu scriptlerde standardised opsiyonel ve zorunlu datacolumnlar vardır ve bunlar her bir müşteri ve proje için özel olarak doldurulur. Bu şekilde sistem genelinde bir structure korunur.

Bu dosyalar bash scriptler aracılığıyla belirli input klasörlerine yönlendirilir ve burada otomatik olarak input parsera atanır. Parser CSV dosyasını okur ve veritabanı kayıtlarını oluşturur.

Müşteriyi kayıt etmek ve projesinin temelini oluşturmak amacıyla 12 tane script dosyası hazırlanıp proje için gerekli temel bilgiler database e kaydedilmiştir. Bu scriptler arasında bağımlılık bulunduğundan yürütme sırasına göre runlanır. İşte o scriptler:

- **01_customer_setup.csv**: customer setup ile database e yeni bir customer eklenir. Bu estup basitce customerin adı, internal sistemde kolaylık olması açısından bir nickname ve yine bir java classı ile üretilen bir hash key ile kaydedilir.
- **02_project_setup.csv**: Project setup ile var olan müşterilere yeni projeler eklenir. Bu projelerin componentleri, template bilgileri, error mail recipienti, üretim için gerekli file amountı ve ultimo regionu csvde eklenir. Örneğin Hintzvv için project setup dosyası şu şekildedir:

- **03_fund_setup.csv**: fund setup dosyası ile müşterinin yeni fundları database e kaydedilir.
- **04_fic_setup.csv**: Fund investment class setup ile fondlara ait olan investment classlara veri bankasına kaydedilir.
- **05_price_series_backfill.csv**:

- `06_insert_input_pattern.csv`: Input pattern setup i ile musteriden gelecek olan dataların runlandığında otomatik olarak ait olduğu input parser tarafından okunabilmesi için bir file input pattern oluşturmaya sağlar.
- `07_staticdatasetup.csv`: Factsheetta bulunan static dataların template'de attribute aracılığıyla kullanılabilmesi için veritabanına kaydedilmesini sağlar. Bu datalar nadir durumlarda değişiklik gösterir örneğin country veya issue fee gibi
- `08_classification_backfill.csv`: Classification backfill ile fon için gerekli temel classification kategori bilgisini kaydeder. Bu classificationlar daha sonrasında çeşitli chartları oluştururken kullanılır. Bu proje için gerekli classificationlar şunlardır:
 - AI Leaders asset classification
 - AI Leaders sector classification
 - AI Leaders country classification
 - AI Leaders currency classification
- `09_benchmark_series_setup.csv`:
- `10_setup_email_list.csv`: Email setup ile factsheet için bir email listesi oluşturuldu. E-Mail- Adressen, welche der Liste hinzugefügt werden, erhalten automatisiert die in Zukunft erzeugten Factsheets.
- `11_map_email_list.csv`: Hier wird die E-Mail-List ID auf die ID des Projektes gemappt.
- `sql_function.csv`:

4.5.2 Services

4.5.3 Document Engine

her bir proje için bir parser bulunur

Input Parser

Charting

Factsheet Parser

4.6 Testing

5 Conclusion

During this internship, I gained both technical and personal development through hands-on work in a professional software development environment. I was able to strengthen my core programming skills, improve my understanding of development workflows, and become more comfortable working in a team setting.

The experience helped me build a solid foundation in Java, databases such as SQL and MongoDB, and essential tools like Git and Linux. In addition, working with tools such as JFreeChart, JasperSoft, and Atlassian products contributed to my practical understanding of real-world software development processes.

Overall, this internship provided valuable insights into both technical development and workplace practices, and it highlighted areas where I can continue to improve, especially in structured learning and documentation usage.

5.1 Learning Outcomes

5.1.1 Soft Skills

- Developed strong self-learning abilities through independent research and problem-solving using real examples.
- Improved communication within a team environment by actively participating in meetings and giving/receiving regular feedback.
- Gained experience in balancing work, study, and personal life in a structured and sustainable way.
- Built a more natural programming mindset by reducing reliance on AI tools and focusing more on understanding fundamentals and reasoning.

5.1.2 Technical Skills

- Acquired a solid foundation in Java through hands-on development tasks.

5 Conclusion

- Gained good working knowledge of SQL and MongoDB, including writing and optimizing queries.
- Strengthened version control practices by regularly using Git in daily workflows.
- Gained practical experience working in a Linux-based development environment.
- Worked with data visualization and reporting tools such as JFreeChart customization and JasperSoft.
- Developed familiarity with Atlassian tools (e.g., Jira, Confluence) in a structured project environment.
- Improved proficiency in Microsoft Excel for data handling and reporting tasks.

5.1.3 Challenges and Areas for Improvement

- Tended to approach learning too quickly and broadly at the beginning, often trying to cover too many topics at once instead of focusing on depth.
- At times, skipped onboarding materials and documentation under the assumption that I was already familiar with the content. For example, I did not review certain Confluence resources (such as the OPSDEV folder) in sufficient detail early on, and only started going through them more thoroughly around week 7 of the internship.
- Clean code, error handling gibi verimli programlamanın ana taslarını teorik olarak bilmekle aktif olarak uygulamak aynı şey olmadıktan pratikte de hep göz önünde bulundurmayı staj sürecimde öğrendim.

This is a test citation Fiig & Isler (2004).

6 AI List

The following table lists the artificial intelligence tools used in this report:

AI Tool	Purpose	Affected Sections
ChatGPT	Text translation and correction	All sections
Claude	LaTeX correction	All sections

Table 6.1: AI tools used in the project

7 Irmak Notes & TODOs

- Hintzvv Ai Leadersin websitesindeki linkini de belgeye ekle, hem goz atmak isteyen bakabilir hem de datenschutztan biraz kaytarmis olursun resmi sitesinde de bulunuyor diye
- benim project managerim seke mi keivn mi? intership projesi icin requirementlari ben kevindan aliyorum ama projeyi bana veren Sekeydi, ve bi yere Ninayi da eklemeyi unutma
- Eger loglarinin oladugunu belirtip Jira Tasks pdfine yonlendirmek istersen ona bir yer bak nere olabilir diye ama muuthis gerekli mi emin degilim
- quellcode vermek yasak, kod vermek istediginde metotlari yuzeysel verebilirsin belki ama direkt kodlari yapistirmek telif hakki gerektiriyormus
- betreuenden Person: telefon ve emailini de ekle
- team eventlerde ops event ve studierenden event icin foto koy



AI LEADERS

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Werbemitteilung

12.06.2026

Anlageziel

Der AI Leaders verfolgt einen zukunftsweisenden Investmentansatz, bei dem Künstliche Intelligenz sowohl das Anlageziel als auch ein zentrales Werkzeug der Fondsverwaltung bildet. Anlageschwerpunkt des Fonds sind Unternehmen, die direkt oder indirekt von künstlicher Intelligenz profitieren. Ziel des Fonds ist die Erwirtschaftung einer aktiennahen Rendite bei möglichst gleichem oder geringerem Risiko. Der global anlegende Aktienfonds investiert bevorzugt in Unternehmen, die zum Erreichen ihrer Ziele auch die Vorteile von Künstlicher Intelligenz nutzen und zu den Leadern ihrer Branche zählen. Eine bedeutsame Rolle bei der Titelauswahl spielt die historische Entwicklung (Bilanzsumme, Gewinnwachstum, Dividendenzahlungen etc.) sowie eine aktuell gute Positionierung in den Hauptgeschäftsbereichen des Unternehmens. Der Fonds investiert dabei entlang der gesamten Wertschöpfungskette der Künstlichen Intelligenz.

Anlagepolitik

Der AI Leaders investiert global in ein breit diversifiziertes Aktienportfolio. Mindestens 51 % des Fondswertes werden in Kapitalbeteiligungen im Sinne des § 2 Absatz 8 des deutschen Investmentsteuergesetzes angelegt. Das Managementteam verwaltet dabei ein Portfolio von bis zu 175 Werten – bestehend aus Herstellern und Anbietern von KI-Hardware und KI-Software

Wertentwicklung

Frühere Wertentwicklung und Kennzahlen sind keine verlässlichen Indikatoren für die künftige Wertentwicklung.



*ab hier Strategieänderung

Wertentwicklung in %

Zeitraum	YTD	1 Jahr	3 Jahr	5 Jahr	Seit Auflage
kumuliert	18,72	25,57	59,31	35,23	106,58
annualisiert	46,54	25,59	16,79	6,22	11,92

Monatliche Wertentwicklung in %

	Jan	Feb	Mrz	Apr	Mai	Jun	Jul	Aug	Sep	Okt	Nov	Dez	YTD
2020	2,14	-4,47	-11,52	14,51	12,45	7,43	4,90	5,28	-1,30	1,02	12,48	5,05	55,39
2021	3,41	-0,70	-9,11	4,53	-4,95	15,76	0,33	6,22	-5,40	7,21	0,90	-5,81	10,27
2022	-17,06	-3,00	-2,14	-10,54	-5,42	-5,64	10,84	-3,64	-7,84	5,58	2,51	-8,41	-38,67
2023	8,04	1,56	3,05	-5,39	11,07	3,17	3,81	-2,42	-3,13	-4,29	10,20	5,52	33,88
2024	3,88	6,39	1,16	-5,13	1,96	5,66	-1,22	-0,26	-0,72	0,82	8,78	0,69	23,44
2025	4,33	-3,50	-10,81	-2,26	6,14	2,85	0,29	-1,15	4,20	5,78	-4,21	-0,10	0,19
2026	0,63	-1,02	-5,98	12,70	13,17	-0,61							18,72

Fondsdetails

ISIN	DE000A2PF0M4
WKN	A2PF0M
Verwaltungsgesellschaft	Monega KAG
Verwahrstelle	Kreissparkasse Köln
Fondsberater	Christian Hintz Vermögensverwaltung GmbH
Geschäftsjahr	01.01 – 31.12
Erstausgabedatum	02.01.2020
Fondswährung	EUR
Minimale Investition	Keine
Mindest-Folgeinvestition	Keine
Fondsvolumen	148,91 Mio. EUR
Ertragsverwendung	ausschüttend

Fondspreise

Rücknahmepreis	199,65 EUR
Ausgabepreis	199,65 EUR

Kosten

Ausgabeaufschlag	0.00%
Laufende Kosten (TER)	1.20%
Performance Fee	Keine



Tilmann Speck
DIPL.-Volkswirt



Christian Hintz
Wirtschaftsberater (WBA)

Über Uns

Die Christian Hintz Vermögensverwaltung GmbH ist ein Wertpapierinstitut nach §15 WpIG. Sie ist seit 1994 in der Vermögensverwaltung und ist eine von 466 zugelassenen Vermögensverwaltungen deutschlandweit. Als Vermögensverwaltung genießt Sie absolute Unabhängigkeit



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KI getütztes Aktienresearch

Das Fundament unserer Strategie ist ein hochmodernes, KI-gestütztes Research-Modell, das die globale Unternehmenslandschaft rund um die Uhr überwacht. Dieses 24/7-Frühwarnsystem prüft ununterbrochen die KI-Relevanz von Unternehmen: Wir identifizieren fortlaufend, welche Player neue Technologien auf den Markt bringen und welche Akteure durch diese Innovationen disruptiert werden könnten. Unsere KI-Agenten durchleuchten die Unternehmen und beurteilen diese anhand der KI-Relevanz, dem Market Standing und anderer KI-Wettbewerber. So erkennen wir Bedrohungen für etablierte Geschäftsmodelle frühzeitig und können Chancen nutzen. Während wir seit 2022 auch im Risikomanagement konsequent auf eine KI setzen, um datengetriebene Stabilität zu gewährleisten und menschliche Fehlentscheidungen zu reduzieren, bleibt die finale Steuerung bewusst in Expertenhand.

KI gestütztes Risikomanagement

Seit September 2022 ist der AI Leaders der erste und bisher einzige Aktienfonds, der ein KI-gestütztes Risikomanagement mittels des TOPAS-Algorithmus von ELAN Capital einsetzt. TOPAS – das „Time-Series Oriented Portfolio Allocation System“ – ist ein moderner, finanzwissenschaftlich gestützter KI-Algorithmus, der für die Portfolio-Allokation, also die ständige Anpassung der Gewichtungen einzelner Portfoliobausteine, konzipiert wurde. Der Algorithmus überwacht und steuert autonom ein im Vorfeld definiertes Risiko-Ertrags-Profil und passt bei Marktveränderungen die Allokation an, um dieses Profil zu jeder Zeit zu erhalten. Das KI-System misst permanent und überwacht 24/7 unter anderem die über 12.000 Korrelationen zwischen den Aktien im Fonds mit dem Ziel, Schwankungen deutlich zu reduzieren. Das Optimierungsverfahren wurde durch das statistische Beratungslabor des Instituts für Statistik der Universität München evaluiert. Mit einer durchschnittlichen Volatilität von rund 15 % liegt der Fonds deutlich unter dem Niveau vergleichbarer KI-Strategien und ermöglicht dennoch die Teilhabe an den Faktorprämien von Wachstumswerten.

Rollierende Volatilität (3 Jahre) ¹



Maximum Drawdown (3 Jahre) ²



Top 10 Positionen

Micron Technology Inc.	5,83%
SK Hynix Inc.	4,23%
Elmos Semiconductor SE	3,39%
NVIDIA Corp.	2,99%
Tesla Inc.	2,11%
Inficon Holding AG	1,94%
ASML Holding N.V.	1,80%
Tokyo Electron Ltd.	1,79%
Broadcom Inc.	1,61%
KLA Corp.	1,58%

Zielmarkt

Der AI Leaders richtet sich an Anleger, die langfristig am Wachstum des Megatrends Künstliche Intelligenz partizipieren möchten und dabei ein verbessertes Risiko-Ertragsprofil gegenüber klassischen Wachstumsfonds anstreben. Durch Investments in KI-getriebene Unternehmen und Geschäftsmodelle können Anleger an der stabilen Weiterentwicklung zahlreicher Branchen wie Healthcare, Automotive oder der industriellen Produktion – angetrieben durch Automatisierung und Robotik – teilhaben. Institutionelle Investoren stellen inzwischen die größte Anlegergruppe des Fonds dar und profitieren von der Kombination aus Growth-Exposure und reduzierter Volatilität. Aufgrund des globalen Aktiencharakters und der damit verbundenen Kursschwankungsrisiken eignet sich der Fonds für Anleger mit einem mittel- bis langfristigen Anlagehorizont und einer entsprechenden Risikotragfähigkeit.

Risikoindikator (SRI)

Normalerweise hohes Risiko aber dafür auch höherer Gewinn

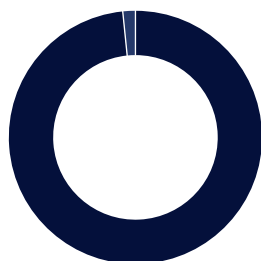
1	2	3	4	5	6	7
---	---	---	---	---	---	---

Normalerweise weniger Risiko aber auch weniger Gewinn

Risikokennzahlen (1 Jahr)

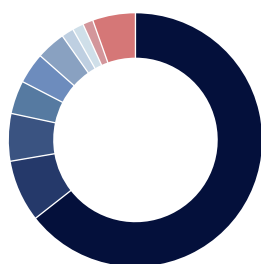
Volatilität	17,15%
Sharpe Ratio	1,39
Max. Drawdown	-10,85%
Time to Recover	24,0 Tage
Höchster Fondspreis	208,61 EUR
Niedrigster Fondspreis	156,51 EUR
Sortino Ratio	1,92
Tracking Error	0,11
Information Ratio	0,13

Asset Allokation



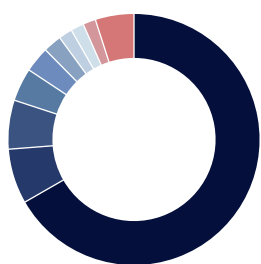
■ Aktien	98,40%
■ Kasse	1,60%

Länder Allokation



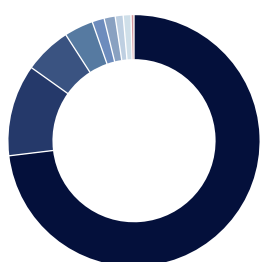
■ USA (einschl. Puerto Rico)	64,42%
■ Bundesrepublik Deutschland	7,88%
■ Japan	6,04%
■ Südkorea	4,23%
■ Niederlande	3,94%
■ Schweiz	3,73%
■ Polen	1,55%
■ Großbritannien und Nordirland	1,42%
■ Israel	1,32%
■ Andere	5,46%

Sektor Allokation



■ Technologie	65,64%
■ Industrie	6,95%
■ Gesundheit / Pharma	6,18%
■ Telekommunikation	4,19%
■ Handel	3,24%
■ Automobil	2,33%
■ Konsumgüter und Dienstleistungen	1,73%
■ Banken	1,65%
■ Medien	1,61%
■ Andere	4,89%

Währungs Allokation



■ USD	73,08%
■ EUR	11,77%
■ JPY	6,04%
■ CHF	3,73%
■ PLN	1,55%
■ GBP	1,42%
■ ILS	1,03%
■ CAD	1,00%
■ HKD	0,38%

Chancen

Global anlegender Aktienfonds mit dem Ziel der Erwirtschaftung einer aktiennahen Rendite bei gleichem oder geringerem Risiko wie der Aktienmarkt.

Bevorzugte Investition in Unternehmen, die zur Erreichung ihrer strategischen Ziele auch die Vorteile künstlicher Intelligenz nutzen und innerhalb ihrer Branche zu führenden Marktteilnehmern zählen.

Titelselektion unter besonderer Berücksichtigung der historischen Entwicklung der Unternehmen in Bezug auf Bilanzsumme, Gewinnwachstum, Dividendenzahlungen und sonstigen Kennziffern sowie der Marktpositionierung in den jeweiligen Hauptgeschäftsbereichen.

Aktives und systematisches Fondsmanagement durch erfahrene Vermögensverwaltung.

Geeignet als Depotbeimischung für Anleger mit langfristigen Anlagehorizont.

Risiken

Aktienrisiko: Aktien unterliegen erfahrungsgemäß starken Kursschwankungen und somit auch dem Risiko von Kursrückgängen.

Zinsänderungsrisiko: Mit der Investition in festverzinsliche Wertpapiere ist das Risiko verbunden, dass sich das Marktzinsniveau während der Haltezeit der Papiere verändert.

Adressenausfallrisiko: Durch den Ausfall eines Ausstellers oder eines Vertragspartners, gegen den der Fonds Ansprüche hat, können für den Fonds Verluste entstehen.

Liquiditätsrisiko: Der Fonds kann einen Teil seines Vermögens in Papieren anlegen, die nicht an einer Börse oder einem ähnlichen Markt gehandelt werden.

Derivaterisiko: Der Fonds setzt Derivate sowohl zu Investitions- als auch zu Absicherungszwecken ein. Die erhöhten Chancen gehen mit erhöhten Verlustrisiken einher.

Zielfondsrisiko: Der Fonds legt in Zielfonds an, um bestimmte Märkte, Regionen oder Themen abzubilden. Die Wertentwicklung einzelner Zielfonds kann hinter der Entwicklung des jeweiligen Marktes zurückbleiben.

Kapitalmarktrisiko: Die Kurs- oder Marktwertentwicklung von Finanzprodukten hängt insbesondere von der Entwicklung der Kapitalmärkte ab.

Wertveränderungsrisiko: Durch die Investitionen des Fondsvermögens in Einzeltitel ergibt sich insbesondere das Risiko von Wertveränderungen der Einzelpositionen.



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Disclaimer

Glossar:

Volatilität¹

Die Volatilität misst den Schwankungsbereich während eines bestimmten Betrachtungszeitraums. Es wird gemessen, wie die Schwankungen eines Fonds über einen bestimmten Zeitraum (z.B. 60 Tage) vom Mittelwert abweichen. Je größer die Schwankungsbreite ist, desto volatil und damit risiko- und chancenreicher ist ein Fonds.

Maximum Drawdown²

Der Maximum Drawdown misst innerhalb eines bestimmten Zeitraums (z.B. 3 Jahre), den zwischenzeitlich größtmöglichen Verlust.

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Kontakt

Christian Hintz Vermögensverwaltung GmbH
Kronprinzstraße 17
70173 Stuttgart

Telefon: 0711 63 22 42/43

Telefax: 0711 25 39 49 59

E-Mail: info@christianhintz.com

Website: www.christianhintz.com

anevis

Certification

This certification is awarded to

Irmak Özdemir

For successfully completing the course

Allgemeine Datenschutzschulung

30/03/2026

Issued Date

30/03/2027

Expiration Date

Jira Tasks

📄 Name	☰ Multi-select	☰ Reporter	# Number	☰ Learned	☰ Live by
Exercise Java Streams	DONE	Sebastian Kempf	1		
Exercise Database	DONE	Sebastian Kempf	2		
Exercise Spring JPA	DONE	Sebastian Kempf	3	follow clean code principles, keep it stupid simple	
Factsheet Project	DONE	Sebastian Kempf	4		
*** BTGPACTUAL-58 make-asset-classification-static	LIVE	Mirco Stapf	5		Nina Thorwart
📌 OPSDEVBOX-651 add region ultimo mapping to the project setup	LIVE	Sebastian Kempf	6	follow design patterns to save data into db and fetch it	Nina Thorwart
MEDICALSTR-27-replace-srri-with-sri-in-template-and-static-data	LIVE	Mirco Stapf	7		Nina Thorwart
OEKOWORLD-7-remove-warning-from-performance-charting	LIVE	Mirco Stapf	8		Nina Thorwart
OEKOWORLD-8 update pc names	LIVE	Mirco Stapf	9		Nina Thorwart
HANSA-29-delete-prices-and-baskets-of-brwag-funds	LIVE	Mirco Stapf	10		Nina Thorwart
SGKB-11-update-pc-names	LIVE	Kevin Reißmann	11		Nina Thorwart
WYDLER-34-clean-up-unused-classifications	LIVE	Mirco Stapf	12		Nina Thorwart
SMARTINV-53-delete-ger-lu-2353395101	LIVE	Mirco Stapf	13		Nina Lawatsch
AQUALUTUM-19-add-fund-name-to-missing-classification-email	LIVE	Mirco Stapf	14		Nina Lawatsch
📌 COMCAP-68-remove-monthly-comment-from-resource-income-fund	LIVE	Kevin Reißmann	15	Webreport parser should be updated in DocEng as well	Nina Thorwart
BANTLEON-233-delete-bantleon-select-corporates	LIVE	Mirco Stapf	16		Nina Thorwart
📌 GSDRCAP-57-rename-customer-to-gsr-d-capital	LIVE	Kevin Reißmann	17	- renaming customer: dont forget to update change_input_pattern.csv - update input folders: uat_create_dirs	Nina Thorwart
BTGPACTUAL-57-switch-off-and-delete-ip-global-strategy	NOT AVAILABLE	Mirco Stapf	18		
HASPA-76-split-currency-allocation-in-two-columns-for-waehrungsfonds	LIVE	Kevin Reißmann	19	exportWith2SplitTable() creates allocation_legend_table_left and right	Nina Thorwart

Aa Name	≡ Multi-select	≡ Reporter	# Number	≡ Learned	≡ Live by
BANTLEON-249-fast-charting-for-bantleon-esg-report	DONE	Katharina Roth Sebastian Kempf	20		
NATIONALB-72-factsheets-for-NB-Short-Term-Euro-I	LIVE	Kevin Reißmann	21		Nina Thorwart
★ HINTZVV-3-factsheets-for-ai-leaders-c	IN TESTING	Kevin Reißmann Stefan Wegele	22		
*** LLBSWISS-255-implement-parser-to-insert-valor	PR	Katharina Roth	23		
OPSDEVBOX-678-make-category-mandatory-in-project-setup-parser	IN TESTING	Sebastian Kempf	24		
OPSDEVBOX-679-fix-marketing-classifications-with-empty-identifiers	LIVE	Sebastian Kempf	25		
OPSDEVBOX-662-move-series-from-definitions-data-to-series-mappings	IN TESTING	Sebastian Kempf	26		
OEKOWORLD-16-add-fund-name-to-email-subject	IN PROGRESS	Mirco Stapf	27		
NATIONALB-76-catch-all-missing-asset-mappings-in-hi-input		Mirco Stapf	28		
SKKBN-28-add-bond-charts-and-figures-to-chance-plus-factsheet	LIVE	Mirco Stapf	29		Nina Thorwart
OPSDEVBOX-680-monitor-dead-projects	IN PROGRESS	Sebastian Kempf	30		
GSPD-6-rename-dws-concept-gs-p-food	IN TESTING	Mirco Stapf	31		

Bibliography

Fiig, T., & Isler, K. (2004). SAS O&D Low Cost Project. In *PODS Consortium Meeting*.
(Minneapolis, MN)

I hereby confirm that:

- I have independently prepared the submitted work, and
- I have not used any additional sources other than those explicitly permitted and indicated in advance

Würzburg, June 24, 2026

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Irmak Damla Özdemir